

CURRICULUM VITAE

Name: Guang Yin
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Position: Associate Professor in Offshore Technology at Department of Mechanical and Structural Engineering and Materials Science, University of Stavanger, Norway

EDUCATION

PhD Tsinghua University, Engineering Mechanics <i>"Prediction of Near-Wall Turbulence and Application in Large Eddy Simulation"</i>	May 2018
BS Tsinghua University, Engineering Mechanics Tsien (Hsue-Shen Tsien) Excellence in Education Program	July 2013

RESEARCH EXPERIENCE

Postdoctoral Researcher in Offshore Technology Department of Mechanical and Structural Engineering and Materials Science, University of Stavanger, Norway	2018-2021
Researcher in Offshore Technology Department of Mechanical and Structural Engineering and Materials Science, University of Stavanger, Norway	2021-2025
Associate Professor in Offshore Technology Department of Mechanical and Structural Engineering and Materials Science, University of Stavanger, Norway	2025-Now

RESEARCH INTERESTS

- Ocean Engineering
- Hydrodynamics on Offshore and Subsea Structure
- Marine Computational Fluid Dynamics
- Fluid Mechanics
- Fluid Structure Interaction
- Scour and Sediment Transport
- Reduced Order Modeling
- Machine Learning Applications

Journal Publications

(*: corresponding author)

1. **Yin, G.**, Huang, W.X. and Xu, C.X., 2017. On near-wall turbulence in minimal flow units. *International Journal of Heat and Fluid Flow*, 65, pp.192-199
2. **Yin, G.**, Huang, W.X. and Xu, C.X., 2018. Prediction of near-wall turbulence using minimal flow unit. *Journal of Fluid Mechanics*, 841, pp.654-673.
3. Zhao, F., **Yin, G.***, & Ong, M. C. (2020). Numerical study on flow around partially buried two-dimensional ribs at high Reynolds numbers. *Ocean Engineering*, 198, 106988
4. **Yin, G.**, Andersen, M. and Ong, M.C., 2020. Numerical simulations of flow around two tandem wall-mounted structures at high Reynolds numbers. *Applied Ocean Research*, 99, p.102124.
5. Andersen, M., **Yin, G.*** and Ong, M.C., 2021. Numerical Simulations of Flow Around Wall-Mounted Square and Trapezoidal Structures at High Reynolds Numbers. *Journal of Offshore Mechanics and Arctic Engineering*, 143(1)
6. **Yin, G.**, & Ong, M. C. (2020). On the wake flow behind a sphere in a pipe flow at low Reynolds numbers. *Physics of Fluids*, 32(10), 103605.
7. **Yin, G.**, & Ong, M. C. (2021). Numerical analysis on flow around a wall-mounted square structure using Dynamic Mode Decomposition. *Ocean Engineering*, 223, 108647.
8. Huang, J., **Yin, G.***, Ong, M.C., Myrhaug, D. and Jia, X., 2021. Numerical investigation of scour beneath pipelines subjected to an oscillatory flow condition. *Journal of Marine Science and Engineering*, 9(10), p.1102.
9. **Yin, G.**, Ong, M.C., Lee, J. and Kim, T., 2021. Numerical simulation of oxygen transport in land-based aquaculture tank. *Aquaculture*, p.736973.
10. **Yin, G.**, Nitter, B. and Ong, M.C., 2021. Numerical simulations of turbulent flow through an orifice plate in a pipe. *Journal of Offshore Mechanics and Arctic Engineering*, 143(4).
11. Janocha, M.J., **Yin, G.*** and Ong, M.C., 2021. Modal Analysis of Wake Behind Stationary and Vibrating Cylinders. *Journal of Offshore Mechanics and Arctic Engineering*, 143(4), p.041902
12. Serta, C.P.V., Janocha, M.J., **Yin, G.** and Ong, M.C., 2021. Numerical simulations of flow-induced vibrations of two rigidly coupled cylinders with uneven diameters in the upper transition Reynolds number regime. *Journal of Fluids and Structures*, 105, p.103332.
13. Xing, Y., Janocha, M.J., **Yin, G.** and Ong, M.C., 2021. CFD Investigation on Hydrodynamic Resistance of a Novel Subsea Shuttle Tanker. *Journal of Marine Science and Engineering*, 9(12), p.1411.
14. Ong, M.C., & **Yin, G.**, 2022. On the three-dimensional wake flow behind a normal flat plate. *Physics of Fluids*, 34(1), p.013603.
15. Fraga, V.S., **Yin, G.***, Ong, M.C. and Myrhaug, D., 2022. CFD investigation on scour beneath different configurations of piggyback pipelines under steady current flow. *Coastal Engineering*, 172, p.104060.
16. Serta Fraga, V., **Yin, G.*** and Ong, M.C., 2022. Three-dimensional numerical simulations and proper orthogonal decomposition analysis of flow over different bottom-mounted ribs. *Ships and Offshore Structures*, 17(4), pp.792-827.
17. Janocha, M.J., Ong, M.C. and **Yin, G.***, 2022. Large eddy simulations and modal decomposition analysis of flow past a cylinder subject to flow-induced vibration. *Physics of Fluids*, 34(4), p.045119.

18. Li, W., Zhou, Q., **Yin, G.**, Ong, M.C., Li, G. and Han, F., 2022. Experimental Investigation and Numerical Modeling of Two-Phase Flow Development and Flow-Induced Vibration of a Multi-Plane Subsea Jumper. *Journal of Marine Science and Engineering*, 10(10), p.1334.
19. Nazvanova, A., **Yin, G.*** and Ong, M.C., 2022. Numerical Investigation of Flow around Two Tandem Cylinders in the Upper Transition Reynolds Number Regime Using Modal Analysis. *Journal of Marine Science and Engineering*, 10(10), p.1501.
20. Jurga, A.P., Janocha, M.J., **Yin, G.*** and Ong, M.C., 2022. Numerical Simulations of Turbulent Flow Through a 90-Deg Pipe Bend. *Journal of Offshore Mechanics and Arctic Engineering*, 144(6), p.061801
21. Janocha, M.J., Fabricius, L.P., **Yin, G.*** and Ong, M.C., 2022. Three-Dimensional Large Eddy Simulations and Proper Orthogonal Decomposition Analysis of Flow Around a Flexibly Supported Circular Cylinder. *Journal of Offshore Mechanics and Arctic Engineering*, 144(6), p.060902.
22. **Yin, G.***, Janocha, M.J. and Ong, M.C., 2022. Estimation of Hydrodynamic Forces on Cylinders Undergoing Flow-Induced Vibrations Based on Modal Analysis. *Journal of Offshore Mechanics and Arctic Engineering*, 144(6), p.060904.
23. Li, W., Song, W., **Yin, G.**, Ong, M.C. and Han, F., 2022. Flow regime identification in the subsea jumper based on electrical capacitance tomography and convolution neural network. *Ocean Engineering*, 266, p.113152.
24. Han, F., Liu, Y., Ong, M.C., **Yin, G.**, Li, W. and Wang, Z., 2022. CFD investigation of blind-tee effects on flow mixing mechanism in subsea pipelines. *Engineering Applications of Computational Fluid Mechanics*, 16(1), pp.1395-1419.
25. **Yin, G.**, Ong, M.C. and Zhang, P., 2023. Numerical investigations of pipe flow downstream a flow conditioner with bundle of tubes. *Engineering Applications of Computational Fluid Mechanics*, 17(1), p.e2154850.
26. Yan, X., Chen, C., **Yin, G.**, Ong, M.C., Ma, Y. and Fan, T., 2023. Numerical investigations on nonlinear effects of catenary mooring systems for a 10-MW FOWT in shallow water. *Ocean Engineering*, 276, p.114207.
27. Schnepf, A., Lopez-Pavon, C., Ong, M.C., **Yin, G.** and Johnsen, Ø., 2023. Feasibility study on suspended inter-array power cables between two spar-type offshore wind turbines. *Ocean Engineering*, 277, p.114215.
28. Nazvanova, A., Ong, M.C. and **Yin, G.***, 2023. A data-driven reduced-order model based on long short-term memory neural network for vortex-induced vibrations of a circular cylinder. *Physics of Fluids*, 35(6).
29. Li, W., Li, J., **Yin, G.** and Ong, M.C., 2023. Experimental and Numerical Study on the Slug Characteristics and Flow-Induced Vibration of a Subsea Rigid M-Shaped Jumper. *Applied Sciences*, 13(13), p.7504.
30. Wang, Z., **Yin, G.***, Ong, M.C. and Chen, Y., 2023. Improved delayed detached eddy simulations of flow past an autonomous underwater helicopter. *Physics of Fluids*, 35(7).
31. Wen, X., Ong, M.C. and **Yin, G.***, 2023. On the three-dimensional effects of the water entry of wedges. *Applied Ocean Research*, 138, p.103649.
32. Liu, Y., Han, F., **Yin, G.**, Ong, M.C., Li, W. and Wang, Z., 2023. Numerical investigation of blind-tee effect on flow mixing in double-bend combinations. *Ocean Engineering*, 287, p.115909.

33. Ma, Y., Chen, C., **Yin, G.**, Ong, M.C., Lu, H. and Fan, T., 2023. Design Methodology of Wind Turbine Rotor Models Based on Aerodynamic Thrust and Torque Equivalence. *Journal of Marine Science and Engineering*, 12(1), p.1.
34. Li, G., Li, W., Janocha, M.J., **Yin, G.** and Ong, M.C., 2023. Large eddy simulations of flow past an inclined circular cylinder: Insights into the three-dimensional effect. *Physics of Fluids*, 35(11).
35. Jurga, A.P., Janocha, M.J., Ong, M.C. and **Yin, G.***, 2024. Numerical Investigations of Turbulent Flow Through a 90-Degree Pipe Bend and Honeycomb Straightener. *Journal of Fluids Engineering*, 146(2), p.021307.
36. **Yin, G.***, Ong, M.C. and Ye, N., 2024. Hydrodynamic Forces on a Near-Bottom Pipeline Subject to Wave-Induced Boundary Layer. *Journal of Offshore Mechanics and Arctic Engineering*, 146(2).
37. Ma, Y., **Yin, G.***, Janocha, M.J., Xing, Y. and Ong, M.C., 2024. Numerical Investigation on Near-Bottom Operation of an Extra-Large Freight Submarine. *Journal of Offshore Mechanics and Arctic Engineering*, 146(2).
38. Han, F., Lan, Q., Liu, Y., **Yin, G.**, Ong, M.C., Li, W. and Wang, Z., 2024. Unveiling Turbulent Flow Dynamics in Blind-Tee Pipelines: Enhancing Fluid Mixing in Subsea Pipeline Systems. *Journal of Marine Science and Engineering*, 12(7), p.1199.
39. Li, G., Janocha, M.J., **Yin, G.**, Ong, M.C. and Li, W., 2024. On the importance of angle-dependent hydrodynamic coefficients in the equilibrium configuration analysis of synthetic fiber towing ropes. *Ocean Engineering*, 300, p.117509.
40. **Yin, G.*** and Ong, M.C., 2024. Wake flow behind a transversely rotating sphere confined in a pipe at low Reynolds numbers. *Physics of Fluids*, 36(9).
41. Gu, Y., **Yin, G.***, Janocha, M.J. and Ong, M.C., 2024. Icing Models and Mitigation Methods for Offshore Wind in Cold Climate Regions: A Review. *Marine Energy Research*, 1(1), p.10002.
42. **Yin, G.**, Gorikhin, A., Olsen, C.L., Liu, S. and Ong, M.C., 2024. Numerical simulations of wave-soil-structure interactions for a subsea cover. *Coastal Engineering*, 189, p.104475.
43. **Yin, G.***, Janocha, M.J. and Ong, M.C., 2024. Physics-Informed Neural Networks for Prediction of a Flow-Induced Vibration Cylinder. *Journal of Offshore Mechanics and Arctic Engineering*, 146(6).
44. Han, F., Lan, Q., Ong, M.C., **Yin, G.*** and Li, W., 2024. Experimental investigation on transition mechanisms and modal decomposition analysis of gas–liquid flow patterns in horizontal–vertical elbow. *Physics of Fluids*, 36(12).
45. Wen, X., **Yin, G.**, Liu, T., Ong, M.C., Wang, C. and Wang, K., 2025. An Optimization Framework for Mooring Design of Floating Offshore Wind Turbines Using a Genetic Algorithm based on a Surrogate Model. *Renewable Energy*, p.122807.
46. Ong, M.C. and **Yin, G.**, 2025. Synergizing machine learning with fluid–structure interaction research: An overview of trends and challenges. *OCEAN: Tsinghua University Press*, 1(1), p.9470002.
47. Ong, M.C. and **Yin, G.***, 2025. A decoupled prediction model for hydrodynamic forces on near-wall cylinders in combined wave-current boundary layers. *Applied Ocean Research*, 162, p.104736.
48. **Yin, G.** and Ong, M.C., 2025. A Physics-Informed Machine Learning Framework Combined With a Reduced-Order Model for Efficient Prediction of Unsteady Vortex Shedding. *Journal of Offshore Mechanics and Arctic Engineering*, 147(5), p.051901.
49. **Yin, G.***, Gundersen, S.Ø. and Ong, M.C., 2025. Numerical Investigation on Wave-Induced Boundary Layer Flow over a Near-Wall Pipeline. *Coasts*, 5(4), p.40.

50. **Yin, G.** and Ong, M.C., 2025. A Physics-Informed Neural Network Framework with Reduced Order Model for Predicting Flow-Induced Vibration. *Results in Engineering*, p.107728.
51. Gu, Y., Janocha, M.J., **Yin, G.** and Ong, M.C., 2025. Feasibility study on spar-type floating offshore wind turbine in ice-covered seas. *Results in Engineering*, p.108050.
52. Wen, X., Ong, M.C., **Yin, G.**, Ludvigsen, T. and Oppedal, F., 2025. Comparative analysis of dissolved oxygen distribution in three ocean-based fish cages based on measured metocean data. *Aquaculture*, p.743466.
53. Wen, X., Ong, M.C., **Yin, G.**, Ludvigsen, T. and Oppedal, F., 2025. Influence of Ocean Currents and Stocking Density on the Dissolved Oxygen Distribution Inside Gravity-Type Fish Cages. *Journal of Offshore Mechanics and Arctic Engineering*, 147(6), p.061301.
54. **Yin, G.**, Ong, M.C., Sævik, S., Gjølsteen, J.K.Ø., Giertsen, E. and Ye, N., 2026. A data-driven model for combined current-wave induced boundary layers. *Applied Ocean Research*, 166, p.104877.
55. Sun, Z., Li, H., **Yin, G.** and Ong, M.C., 2026. Numerical investigation of Jet–Roll coupled flow characteristics in an aquaculture vessel tank. *Ocean Engineering*, 343, p.123264.
56. Le, M.J., Wang, S., **Yin, G.** and Ong, M.C., 2026. A Computational Fluid Dynamics-Based Parametric Study of Trimaran Form Factor and Resistance Characteristics. *Journal of Offshore Mechanics and Arctic Engineering*, 148(1), p.011903.
57. Li, G., Ye, H., Li, W., **Yin, G.** and Ong, M.C., 2026. Snap Load Dynamics in Non-Metallic Armored Umbilical Cables under Heave Excitation. *Results in Engineering*, p.109689.
58. Sun, Z., Li, H., **Yin, G.** and Ong, M.C., 2026. Numerical simulation of dissolved oxygen transport under jet–roll coupled hydrodynamics in an aquaculture vessel tank. *Ocean Engineering*, 350, p.124240.
59. Wang, S., Le, M.J., **Yin, G.** and Ong, M.C., 2026. A prediction methodology of trimaran form factor (1+k) based on neural network. *Ocean Engineering*, 351, p.124446.
60. Zhao, P.Q., **Yin, G.*** and Ong, M.C., 2026. Separable physics-informed neural networks for modelling unsteady flow around a circular cylinder near the plane wall. *Ocean Engineering*, 359, p.125984.
61. Li, Y., Wen, X., **Yin, G.**, Jia, L., Huang, J., Ye, N. and Ong, M.C., 2026. Numerical Investigation of Coupled Vortex-Induced Vibration and Scour of Subsea Pipelines under Ocean Currents. *Journal of Marine Science and Engineering*, 14(14), p.1279.
62. Li, G., **Yin, G.***, Li, W. and Ong, M.C., 2026. Hydrodynamic force and vortex dynamics around an inclined cylinder in oscillatory flow. *Physics of Fluids*, 38(8).

Conference Papers

1. Taibi, R., **Yin, G.** and Ong, M.C., 2021, November. CFD investigation of internal elbow pipe flows in laminar regime. In *IOP Conference Series: Materials Science and Engineering* (Vol. 1201, No. 1, p. 012012). IOP Publishing.
2. Jurga, A.P., Janocha, M.J., **Yin, G.**, Giljarhus, K.E.T. and Ong, M.C., 2021, November. Validation and assessment of different RANS turbulence models for simulating turbulent flow through an orifice plate. In *IOP Conference Series: Materials Science and Engineering* (Vol. 1201, No. 1, p. 012019). IOP Publishing.
3. **Yin, G.**, Zhang, Y. and Ong, M.C., 2021, November. Numerical investigations of flow around subsea covers at high Reynolds numbers. In *IOP Conference Series: Materials Science and Engineering* (Vol. 1201, No. 1, p. 012013). IOP Publishing.

4. Wan, J., **Yin, G.**, Liu, S., Ong, M.C. and Moen, P., 2021, June. Numerical Simulations of Flows Around Subsea Cover. In International Conference on Offshore Mechanics and Arctic Engineering (Vol. 85161, p. V006T06A053). American Society of Mechanical Engineers.
5. Huang, J., **Yin, G.**, Ong, M.C. and Jia, X., 2020, August. Numerical Investigation of Scour Beneath a Subsea Piggyback Pipeline. In International Conference on Offshore Mechanics and Arctic Engineering (Vol. 84379, p. V06AT06A003). American Society of Mechanical Engineers.
6. **Yin, G.**, Monaci, T. and Ong, M.C., 2019, November. Numerical simulation of flow around two 5: 1 rectangular cylinders at a high Reynolds Number. In IOP Conference Series: Materials Science and Engineering (Vol. 700, No. 1, p. 012010). IOP Publishing.
7. **Yin, G.**, Cheng, Z., Liu, S. and Ong, M.C., 2019, June. Numerical Investigation of Scour around Subsea Pipelines near the Seabed. In International Conference on Offshore Mechanics and Arctic Engineering (Vol. 58769, p. V001T01A005). American Society of Mechanical Engineers.

PROJECT

- Horizon Europe ERC-Starting Grant 2024: Sparse Measurement-Based Reduced-Order Models of Fluid-Structure-Interactions (SMARTFLUIDS, Project ID: 101165107)
Role: Principal Investigator
- Smart Aquaculture Research Centre hosted by Chonnam National University, Korea
Role: Key Researcher
- The Research Council of Norway: Innovation Project for the Industrial Sector - Subsea-Resident Autonomous ROV with a Minimal Environmental Footprint in cooperation with IKM Subsea AS, Norway
Role: Key Researcher
- JIP Dynamic Response of Offshore Pipelines on the Seabed (DROPS) coordinated by SINTEF Ocean AS, Norway
Role: Key Researcher
- Numerical Investigations on Pipe-Soil-Fluid Interaction in Three-Dimensions for Pipes Resting on Uneven Seabed in cooperation with SINTEF Ocean AS, Norway and CNOOC Research Institute Ltd, China
Role: Key Researcher

INVITED PRESENTATIONS

- “Three-dimensional effects in bluff body wake flows”, at State Key Laboratory of Hydraulic Engineering Simulation and Safety, Tianjin University, Tianjin, Jun 29th, 2022.
- “Research Activities on Marine Hydrodynamics at University of Stavanger”, at Institution of Fluid Mechanics, Department of Engineering Mechanics, Tsinghua University, Beijing, December 25th, 2024.

- “Numerical Investigation of Scour beneath Pipelines”, at Fisheries College, Ocean University of China, Qingdao, December 30th, 2024.
- “Synergizing machine learning with fluid-structure interaction research”, at Delft University of Technology, March 3rd, 2026.

SUPERVISION EXPERIENCE

- Number of Ph.D. Students: 4
- Number of Master Students: 14

TEACHING EXPERIENCE

- OFF925 Marine and Subsea Technology Project Course (10 credits) (University of Stavanger)
Guest lecturing in Autumn 2020, Spring 2022.
Topics: Basic equations of turbulence, Direct Numerical Simulations, Reynolds-Averaged Navier-Stokes equations Simulations, Large Eddy Simulations, wall function.
- OFF900 Environmental Loads on Structures (10 credits) (University of Stavanger)
Guest lecturing in Autumn 2025.
Topics: Regular and Irregular Waves, Stochastic description of environmental loads.
- OFF580 Marine Technology and Design (10 credits) (University of Stavanger)
Guest lecturing in Spring 2024, Spring 2025.
Topics: Linear and nonlinear waves, rigid body dynamics, hydrodynamic loads on floating structures, equations of motions for floating structures.
- OFF505 Oceanography (5 credits) (University of Stavanger)
Lecturing in Autumn 2023, Autumn 2023.
Topics: Waves, atmospheric circulation, atmospheric boundary layer and wind, ocean currents.
- OFF600 Marine Operation (10 credits) (University of Stavanger)
Guest lecturing in Autumn 2024.
Topics: Crane tip lifting operation.

PROFESSIONAL SERVICE

- Session Chairman for International Conference on Ocean, Offshore and Arctic Engineering (OMAE 2020, 2021, 2022, 2023, 2024), ASME.
- Organizer of 1st Marine Computational Fluid Dynamics Seminar (Marine CFD). see <https://www.uis.no/en/information-technology/1st-marine-computational-fluid-dynamics-seminar-marine-cfd>

Peer-Reviewed Activities:

Regular reviewer for journals of Physics of Fluid; Journal of Fluids and Structure; Marine Structures: No. of assignments; Ocean Engineering; Coastal Engineering and International Conference on Ocean, Offshore and Arctic Engineering (OMAE), ASME.

HONORS AND AWARDS

- Cover article and featured article of Physics of Fluids “Yin, G., & Ong, M. C. (2020). On the wake flow behind a sphere in a pipe flow at low Reynolds numbers. Physics of Fluids, 32(10), 103605.”
- Featured article of Physics of Fluids “Wang, Z., Yin, G., Ong, M.C. and Chen, Y., 2023. Improved delayed detached eddy simulations of flow past an autonomous underwater helicopter. Physics of Fluids, 35(7).”
- Cover article and featured article of Journal of Marine Science and Engineering “Nazvanova, A., Yin, G. and Ong, M.C., 2022. Numerical Investigation of Flow around Two Tandem Cylinders in the Upper Transition Reynolds Number Regime Using Modal Analysis. Journal of Marine Science and Engineering, 10(10), p.1501.”
- Outreach Scholarship 38th OMAE in Glasgow, Scotland, UK, ASME
- Outstanding Academic Papers by Students of Tsinghua University (2013)
- Twice Tsinghua Xuetang Talents Program Scholarship (2010~2012)
- Tsinghua's Friend- Zheng Geru Scholarship (2010~2011)
- Meritorious Winner of Mathematical Contest In Modeling (MCM, 2011)
- First Prize of The Chinese Mathematics Competitions Beijing District (2010)